Hacettepe University
Department of Industrial Engineering

EMÜ414- Mathematical Programming and Computer Applications

2025-2026 Spring

Term Project – First Deliverable

Deadline: April 28, 2026, 23:59

1. Team Formation

The project must be completed in teams of 4 to 6 students.

2. Project Topic Selection

Each team must select a project topic that involves an operations research problem requiring optimization-based decision making.

You may choose one of the following three project types. The difference between them is mainly the primary focus of the project.

> **a. Research-oriented operations research problem.** The main focus is on the mathematical model and solution methodology. Practical realism is not the primary concern. The problem may be hypothetical or a simplified version of a real problem, provided that it is rich enough to allow meaningful modeling and computational analysis.
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> **b. Application of operations research methods to a real problem.** The main focus is on solving an actual real-world problem using operations research methods. The project should remain as close to reality as possible. Simplifications should be limited and clearly stated. Teams are expected to explain the practical setting and, where possible, compare the current system with the proposed solution.
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> **c. Operations research case study.** The main focus is on presenting and analyzing a specific practical decision situation in a structured and self-contained way. A case study should involve a sufficiently complex real or realistic problem, together with the relevant data, modeling approach, computational results, and discussion. Compared with the other two categories, a case study places more emphasis on the overall problem context, data, and discussion of results.

In all three categories, the selected problem should be suitable for the development of a mathematical optimization model and a Gurobi-based solution in the later stages of the project.

You may build on a problem studied in the literature or adapt a real-world problem, but direct copying of an existing study or model is not acceptable. Your project should reflect your team's own understanding and development of the problem.

3. First Deliverable Requirements

The first deliverable is intended as a short project proposal. Since this is an initial submission, the report should be concise and focused.

Each team must submit a 4-page PDF report produced in LaTeX. The report must include the following:

- team name
- team members
- an introduction to the problem
- a description of the decisions to be made
- the objective to be improved
- the factors that restrict these decisions
- the main assumptions, if any
- the classification of the problem within well-known operations research problem classes, such as routing, scheduling, facility location, network flow, or similar categories
- a brief description of the anticipated mathematical model structure (for example, whether the model is expected to be linear or nonlinear, and continuous, integer, or mixed-integer)

4. Report Preparation

The report must be prepared in LaTeX.

You are strongly encouraged to use Overleaf, but any other LaTeX platform is also acceptable.

5. Submission

Submit your first deliverable electronically through www.hadi.hacettepe.edu.tr .

Do not submit a printed copy.

Your submission must include:

- the 2-page PDF report
- the complete raw LaTeX project files
- any other documents that is required to understand your project

The submission deadline is April 28, 2026, at 23:59.